The Long-run Effects of Transportation Productivity on the US economy

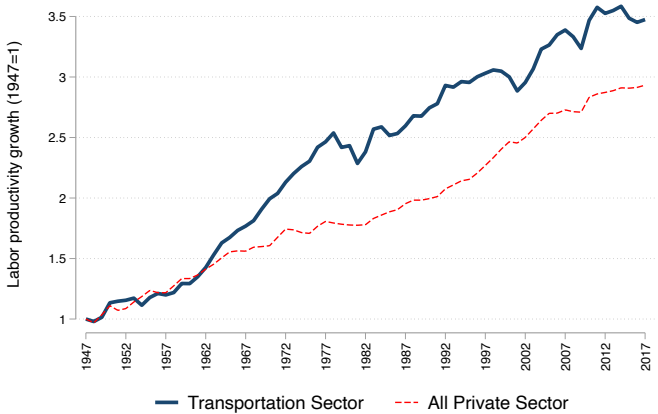
Kerem Coşar
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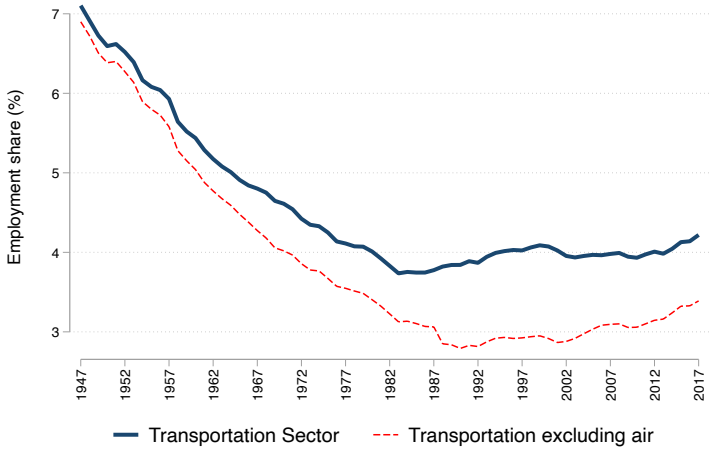
October 2026

US Labor productivity in Transportation



Notes: US real gross labor productivity of transportation sector vs all other private industries; normalized to 1 in 1947, computed from BEA data.

US Transportation employment share



Notes: Authors' calculation from BEA data.

Objective of the paper

The effects of transportation productivity on

1. **aggregate**: real income
2. **spatial**: geography of economic activity
3. **sectoral**: direct and indirect

We build a multi-region & multi-sector model with

- an explicit transportation sector (freight)
- complementarity between transportation services & tradable products
- input-output and investment linkages

Quantify these effects for the US economy over the period 1947-2017.

Main results

With higher transportation productivity (1947-2017),

- US aggregate welfare = + 3.3% $\Rightarrow$ 2.3 times more than implied by sectoral share.
  - Interaction of input-output linkages and complementarity
  - Labor productivity: higher increase in sectors that are heavy users of transportation services (mining, agriculture,...)
- Activity shifts away from the East
  - towards remote regions (w/ high transportation margins)

Related literature

- Literature on infrastructure and transportation costs

  $\Rightarrow$ beyond infrastructure, explicit transportation sector

- G.E. models with an explicit transportation sector
  - Falvey (1976), Casas (1983), Casas and Choi (1985)
  - Behrens and Picard (2011), Ishikawa and Tarui (2018), Wong (2020)
  - Adamopoulos (2011), Herrendorf, Schmitz and Teixeira (2012)

  $\Rightarrow$ quantitative multi-region, multi-sector model

- Multi-region or multi-sector models
  - Jones (2011), Baqaee and Farhi (2022), Osotimehin and Popov (2023), ...
  - Caliendo, Parro, Rossi-Hansberg, Sarte (2017), Baqaee and Farhi (2024)

  $\Rightarrow$ focus on transportation (spatial linkages + capital accumulation)

Model

Model

- Regions and sectors:
  - ℓ regions (index: o, d)
  - n sectors (index: i, j) + transportation sector
- Perfect competition
- Labor mobile across regions & sectors
- Region-sector specific capital accumulation

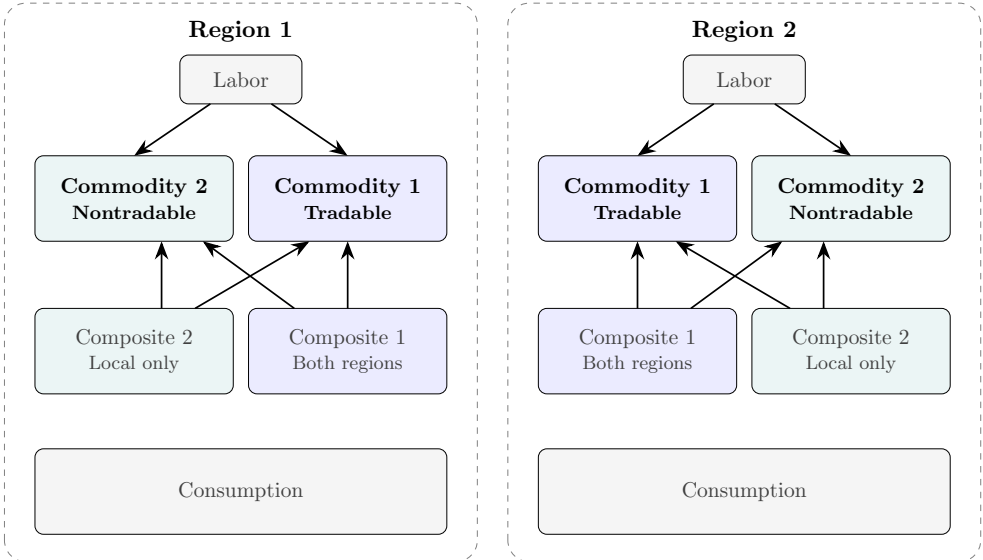
Production structure

Each region produces, in each sector

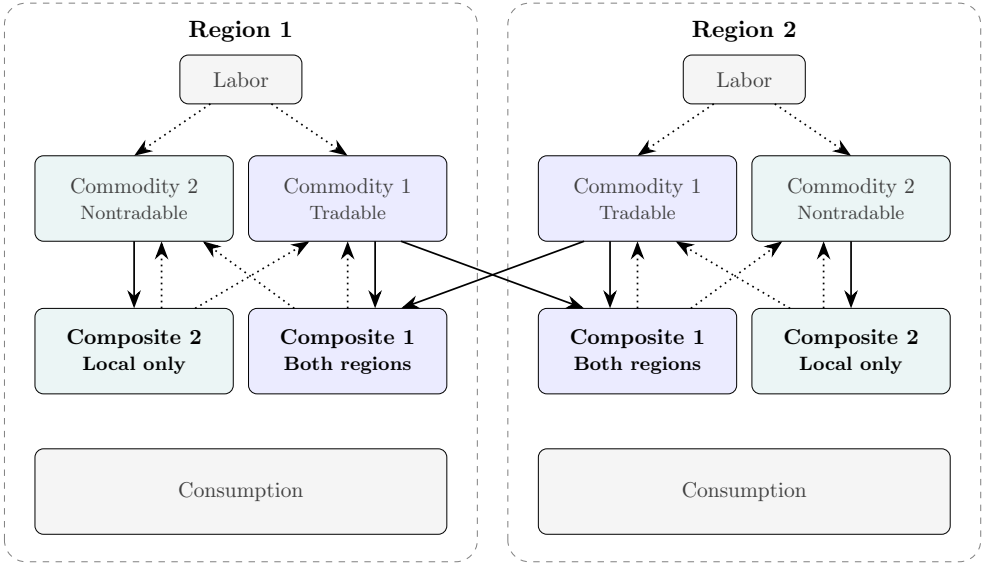
1) A local variety = commodity
   using labor, capital, and intermediate goods.
   tradable or nontradable

2) A local composite product
   using local varieties (commodities)
   nontradable
   used for local consumption and investment, and as intermediate goods

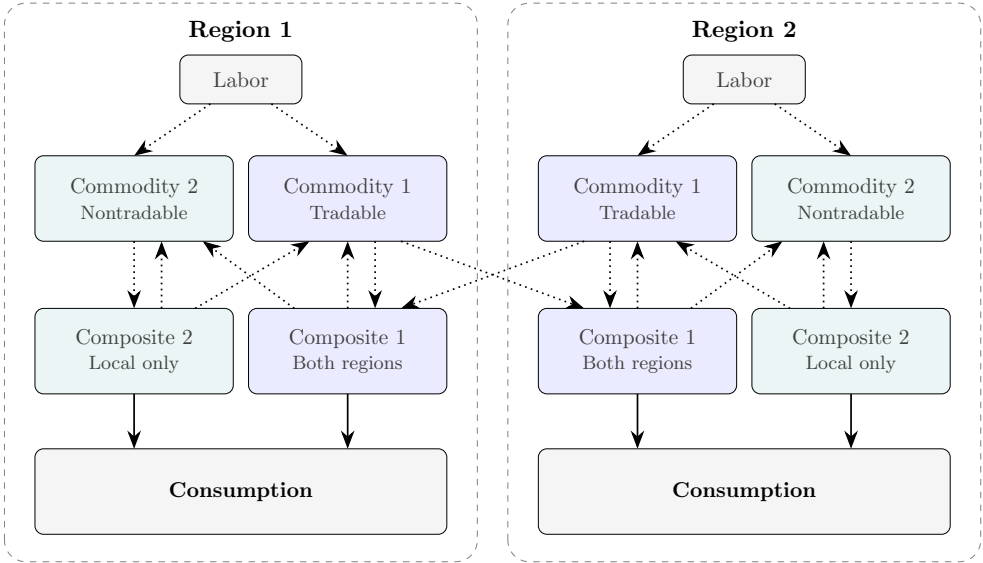
Simplified model (2 regions, no K)



Simplified model (2 regions, no K)



Simplified model (2 regions, no K)



Sector-region commmodity

Representative firm for sector i in region o

$$Q_{io} = \left[(1-\alpha_i)^{1-\sigma} (B_{io} K_{io}^{\theta_i} L_{io}^{1-\theta_i})^{\sigma} + \alpha_i^{1-\sigma} \left(\sum_{j=1}^{n} v_{ij}^{1-\rho} \tilde{X}_{io,jo}^{\rho} \right)^{\frac{\sigma}{\rho}} \right]^{\frac{1}{\sigma}}$$

Functional form for all sectors, including transportation, which has a nationwide productivity B_T

Local composite products

- Local composite product i in region d

  For tradable products = CES aggregate of regional varieties with an Armington elasticity: $1/(1-\gamma) > 1$.

$$\tilde{Q}_{id} = \left(\sum_{o=1}^{\ell} \left(\frac{1}{\ell} \right)^{1-\gamma} \phi_{iod}^{1-\gamma} Q_{iod}^{\gamma} \right)^{\frac{1}{\gamma}}$$

- Used locally for consumption and investment, and as an intermediate input.

Transportation

- We abstract from direct final use of transportation (passenger transportation).
- Transportation services are complementary to trading Armington varieties.
- Each unit of tradable product i shipped from region o to region d (including $o = d$) requires t_{iod} unit of transportation services sourced from the origin:

  price of variety io in region d:

$$p_{iod} = p_{io} + t_{iod} \cdot p_{T,o} = p_{io} + \bar{t}_i \cdot \text{distance}_{od} \cdot p_{T,o}$$

- For tradable services $\Rightarrow t_{iod} = 0$
  For non-tradable services $\Rightarrow t_{idd} = 0$ and $t_{iod} = +\infty, \quad o \neq d$

Final demand and capital accumulation

Per capita consumption: $$c_o = \left(\sum_{i=1}^{n} \xi_i^{1-\chi} \tilde{c}_{io}^{\chi} \right)^{1/\chi}$$

Investment: $$I_{io} = \left(\sum_{j=1}^{n} \eta_{ij}^{1-\chi_I} \tilde{I}_{io,jo}^{\chi_I} \right)^{1/\chi_I}$$

Capital: $$K'_{io} = (1 - \delta_i) K_{io} + I_{io}$$

Market clearing

- Transportation: $Q_{transp,o} = \sum_{i=1}^{n} \sum_{d=1}^{\ell} Q_{iod} \cdot t_{iod}$
- Local composite: $\tilde{Q}_{io} = L_{io}\tilde{c}_{io} + \sum_{k=1}^{n+1} \tilde{I}_{ko,io} + \sum_{k=1}^{n+1} \tilde{X}_{ko,io}$
- Local variety: $Q_{io} = \sum_{d=1}^{\ell} Q_{iod}$
- Labor: $\sum_{o=1}^{\ell} \sum_{i=1}^{n+1} L_{io} = \overline{L}$

Spatial equilibrium with capital accumulation

Tractable dynamic model with local capital stock (Kleinman et al's, 2023)

▸ Households set-up

- Workers
  - mobile
  - hand-to-mouth
- Capitalists
  - immobile
  - intertemporal optimization

We focus on the steady-state equilibrium with real wage equalization across regions.

▸ Solution algorithm

Calibration

Calibration

US economy in 2017

- $n = 6$ sectors + transp. ▸ details services ▸ details manuf
  - agriculture, mining
  - heavy & light manufacturing.
  - tradable and non-tradable services
- $\ell = 18$ regions, sub-regions of 9 Census divisions ▸ map
- Key elasticities from the literature ▸ elasticities
  - Armington elasticity between regional varieties $= 8$.

Calibration: Baseline

Parameters $(t_i, \xi_i, \bar{\phi}_i, \alpha_i, v_{ij}, B_{io}, B_T, \eta_{ij}, \theta_i)$ calibrated jointly to match US 2017 data on...

- Share of region o in sector i's total labor
- Share of intermediate inputs used by each sector (IO table with TSA)
- Share of capital inputs used by each sector (vom Lehn and Winberry, 2022)
- Labor share of income
- Sectoral value added shares (recomputed after removing final use of transport)
- Value added to gross output ratio in the transportation sector
- Distance elasticities and out-of-region coefficients, for tradable sectors (CFS-FAF):

$$\underbrace{\ln Q_{iod}}_{\text{trade flows}} = \varphi_{1i} \cdot \ln(dist_{od}) + \varphi_{2i} \mathbb{1}_{o \neq d} + \mu_{oi} + \mu_{di} + \epsilon_{iod}$$

▸ Calibration targets vs data

Results

Counterfactual analysis

- Counterfactual economy: what if B_T had stayed at its past value?
- Compute $B_{T,1963}/B_{T,2017}$ and $B_{T,1947}/B_{T,2017}$ using Solow residual approach
- Compute resulting changes in outcomes relative to the counterfactual value (gain due to increase in B_T).

External validity: interregional trade

We digitized the 1963 U.S. Census of Transportation (3 dgt NAICS **manufacturing** industries; 9 Census divisions) details

In line with empirical evidence, model-predicted trade flows less sensitive to distance when transportation costs are lower.

	Data	Model
$\ln(dist)$	-1.198	-1.613
	(0.000)	(0.000)
$\ln(dist) \times \mathcal{I}(\text{year} = 2017)$	0.167	0.163
	(0.055)	(0.000)
N	324	324
Adjusted within R^2	0.854	0.963
Change in distance elasticity	-14%	-10.1%

Welfare impact

We compute the change in outcome variables relative to the counterfactual.

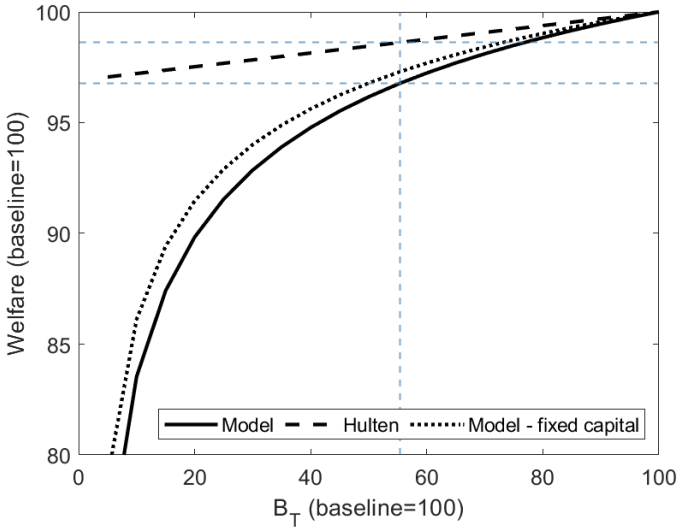
$$\Delta B_T = +81\% \quad \Rightarrow \quad \Delta \mathrm{C} = +3.3\%$$

Is that a large effect?

- Much larger effect ($\times 2.3$) than implied by its sectoral share (VA/GDP = 3%)

$$\text{Hulten (1978):} \quad \Delta \mathrm{C} = \frac{VA_i}{GDP} \cdot \Delta B_i$$

Nonlinear effect



A decomposition of the three main amplification channels

- 3 main amplification channel : transp. complementarity, IO linkages, capital
- To obtain a decomposition, we compute
  - 8 variants of the model, obtained by switching each channel on or off, recalibrating the model each time.
  - how ΔC changes after removing each channel when the other two are on or off.
  - the weighted average of each effect (with Shapley weights)

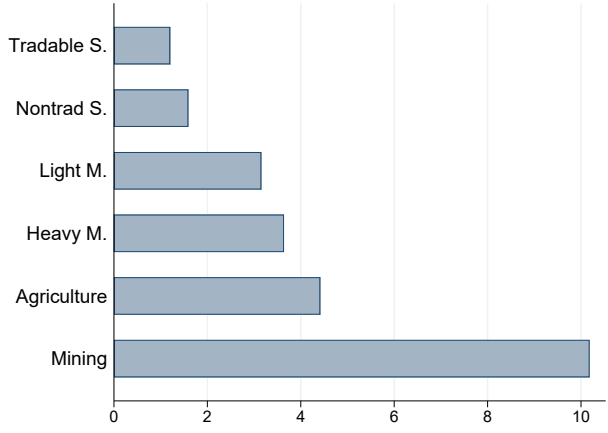
A decomposition of the three main amplification channels

	First-out	Last-out	Average
Transp. complementarity	0.206	0.190	0.204
IO linkages	0.260	0.145	0.208
Capital	0.747	0.417	0.588

- Transportation complementarity + IO explain 40% of the total effect

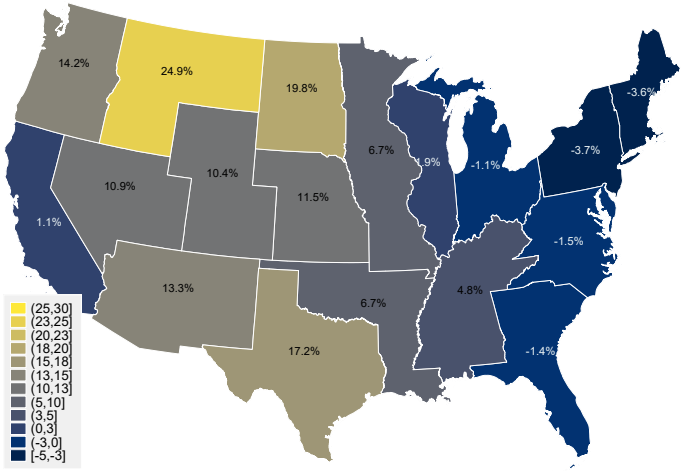
Sectoral productivity changes (%)

Larger labor productivity increases in sectors that are transportation intensive:



Regional output changes (%)

Economic activity shifts towards West while North East and South Atlantic regions shrink:



Determinants of regional changes

Gross output increases the most in regions with high transport margins (κ), inducing increased specialization along comparative advantages.

	Agriculture	Mining	Heavy M.	Light M.	Tradable S.	Nontrad. S.
$\text{corr}(dQ/Q, \kappa)$	0.94	0.98	0.96	0.93	-	-
	(1.1×10^{-8})	(1.9×10^{-12})	(1.6×10^{-10})	(2.5×10^{-8})	-	-
$\text{corr}(dQ/Q, B)$	0.21	0.31	0.18	0.30	-0.04	0.003
	(4.2×10^{-6})	(1.2×10^{-7})	(2×10^{-4})	(1×10^{-4})	(0.40)	(0.38)

Notes: The last line shows partial correlation, controlling for regional population changes. Transportation margin is the ratio of freight costs to the value of the product. p-value in parentheses.

Comparison with iceberg specification

Iceberg costs:

- Transp costs are proportional to the value of the product
  Purchaser price of product $p_{iod} = \tau_{iod}\, p_{io}$ with $\tau_{iod} = (distance_{od})^{\zeta_i}$
  instead of $p_{iod} = p_{io} + t_{iod} \cdot p_{To}$,
- Counterfactual: uniform change in the scale of τ to match the change in the share of transp value added.

Summary of results ▸ Details

- Aggregate effect is smaller ($\Delta C = 3\%$ and multiplier$= 2.1$)
- Transp margins no longer key determinant of gross output changes
- No change in the sensitivity of trade to distance (which depend only on parameters γ and ζ_i)

Summary

- Transportation role in GDP $> \times 2$ effect implied by the sector's size
- Strong propagation through capital accumulation and interaction between sectoral linkages and nonlinearities

A useful framework on a topic highly relevant for many countries

- Amplification effect likely to be stronger for developing countries
- Going beyond the evaluation of infrastructure projects and confronting key statistics of the transport sector.
- Careful treatment of transportation costs key for the quantification

APPENDIX

Households

Hand-to-mouth workers

- Budget constraint

$$\sum_{i=1}^{n} \tilde{p}_{id} \tilde{c}^{\mathrm{w}}_{id} = w_d$$

Capitalists

- Intertemporal utility

$$\sum_{t=0}^{\infty} \beta^t \log c^k_{dt},$$

- Budget constraint

$$\sum_{i=1}^{n} \tilde{p}_{id} \tilde{c}^{\mathrm{k}}_{id} + p^k_{jd,t} k_{jd,t+1} = (r_{jd,t} + 1 - \delta_j) p^k_{jd,t} k_{jd,t}$$

with per-period consumption (both for workers and capitalists): $c_d = \left(\sum_{i=1}^{n} \xi_i^{1-\chi} \tilde{c}_{id}^{\chi} \right)^{\frac{1}{\chi}}$

back

Equilibrium prices

- Composite-product prices

$$\tilde{p}_{id} = \begin{cases} \left[\sum_{o=1}^{\ell} \frac{1}{\ell} \phi_{iod} (p_{io} + t_{iod} \cdot p_{transp,o})^{-\frac{\gamma}{1-\gamma}}\right]^{-\frac{1-\gamma}{\gamma}} & i \in \mathcal{P}_{\mathcal{T}} \\ p_{id} & i \in \mathcal{P}_{\mathcal{NT}} \end{cases}$$

- Local variety prices

$$p_{io} = A_{io}^{-1} \left[(1-\alpha_i) q_{io}^{-\frac{\sigma}{1-\sigma}} B_{io}^{\frac{\sigma}{1-\sigma}} + \alpha_i \left[\sum_{j=1}^{n} v_{ij} \tilde{p}_{jo}^{-\frac{\rho}{1-\rho}} \right]^{\frac{1-\rho}{\rho} \frac{\sigma}{1-\sigma}} \right]^{-\frac{1-\sigma}{\sigma}}$$

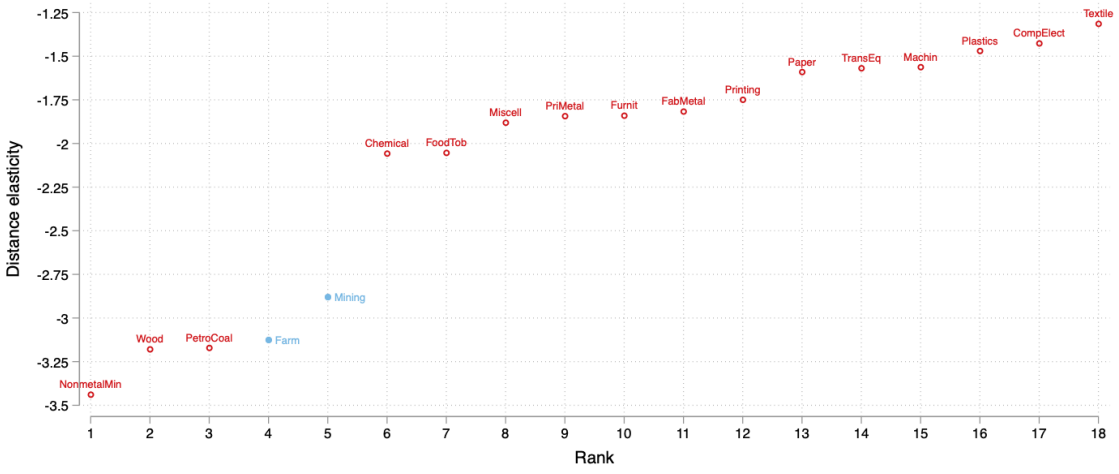
- Capital-labor composite

$$q_{io} = [(r + \delta_i) p_{io}^k]^{\theta_i} w_o^{1-\theta_i}$$

back

Transportation intensity of tradable industries

Back



Classification of industries, tradable services

BACK

Jensen and Kletzer (2005):

- Wholesale trade
- Rental and leasing services and lessors of intangible assets
- Legal services
- Computer systems design and related services
- Miscellaneous professional, scientific, and technical services
- Management of companies and enterprises
- Publishing industries, except internet (includes software)
- Motion picture and sound recording industries
- Broadcasting and telecommunications
- Data processing, internet publishing, and other information services
- Federal Reserve banks, credit intermediation, and related activities
- Securities, commodity contracts, and investments
- Insurance carriers and related activities
- Funds, trusts, and other financial vehicles

Classification of industries, non-tradable services

BACK

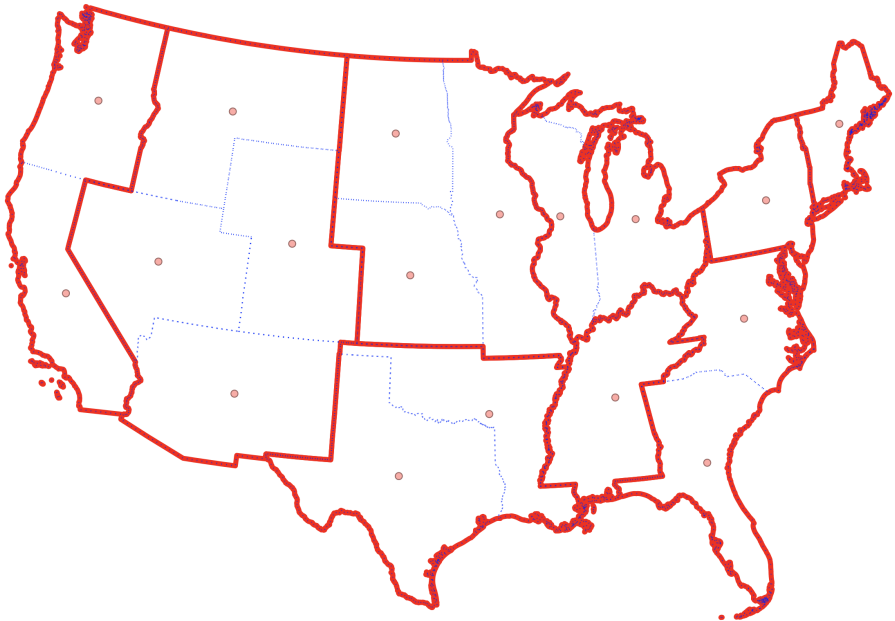
Jensen and Kletzer (2005):

- Retail trade
- Utilities
- Construction
- Real estate
- Administrative and support services
- Waste management and remediation services
- Educational services
- Ambulatory health care services
- Hospitals and nursing and residential care facilities
- Social assistance
- Performing arts, spectator sports, museums, and related activities
- Amusements, gambling, and recreation industries
- Accommodation
- Food services and drinking places
- Other services, except government

Back

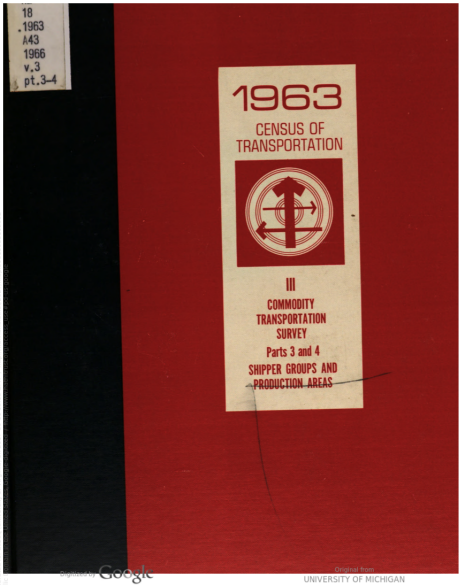

PacNW
MOUNT1
WNC3
NewEng
WNC2
MidAtl
ENC1
ENC2
MOUNT2
MOUNT3
WNC1
Cal
SA2
WSC
MOUNT4
ESC
SA1
Texas

Back



1963 Census of Transportation

Back



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Original from
UNIVERSITY OF MICHIGAN

TABLE 9. **Tons and Ton-Miles of Shipments—Geographic Division of Destination and of Origin: 1963**

Geographic division of origin	Tons and ton-miles	Percent distribution by geographic division of destination									
		All divisions	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific[1]
TONS OF SHIPMENTS	(Thousands of tons)										
United States, total	39,787	100.0	5.6	16.9	24.8	18.2	14.0	4.3	6.4	1.8	8.0
Shipments originating from—											
New England	1,069	100.0	56.5	36.7	-	-	6.8	-	-	-	-
Middle Atlantic	1,986	100.0	2.2	72.5	5.9	1.9	17.2	-	.3	-	-
East North Central	7,530	100.0	2.8	10.2	70.9	2.7	7.3	4.2	1.1	.1	.7
West North Central	17,228	100.0	6.3	14.8	18.7	39.2	8.2	2.0	5.9	1.0	3.9
South Atlantic	4,053	100.0	2.1	26.3	5.4	1.0	63.7	1.4	.1	-	-
East South Central	2,265	100.0	4.3	2.8	24.1	2.5	14.9	41.2	8.7	-	1.5
West South Central	2,285	100.0	3.0	12.2	6.6	4.6	4.5	1.6	52.3	12.0	3.2
Mountain	893	100.0	5.8	17.8	17.2	3.6	8.0	-	5.4	22.5	19.7
Pacific[2]	2,478	100.0	-	.1	4.4	.2	3.4	1.7	-	2.8	87.4
TON-MILES	(Millions of ton-miles)										
United States, total	15,910	100.0	11.3	27.3	15.7	5.1	17.1	3.2	6.1	3.0	11.2
Shipments originating from—											
New England	154	100.0	30.7	44.1	-	-	25.2	-	-	-	-
Middle Atlantic	289	100.0	3.2	46.9	13.6	11.6	22.1	-	2.6	-	-
East North Central	1,762	100.0	9.6	23.6	29.8	3.0	17.2	6.2	4.1	.7	5.8
West North Central	8,823	100.0	14.3	28.8	12.7	6.9	15.5	2.3	6.7	1.7	11.1
South Atlantic	1,120	100.0	2.5	44.0	9.6	2.4	39.9	1.0	.2	-	.4
East South Central	798	100.0	12.3	4.8	33.9	5.2	16.6	12.3	6.9	-	8.0
West South Central	1,276	100.0	8.2	29.6	8.6	1.9	7.4	.5	16.5	19.4	7.9
Mountain	817	100.0	11.1	31.3	17.4	2.6	13.9	-	5.4	3.9	14.4
Pacific[2]	911	100.0	-	.7	21.7	.6	18.0	8.8	.1	3.8	46.3

[1]Shipments to Alaska and Hawaii from the 48 conterminous States and the District of Columbia are included.
[2]No data were obtained on shipments originating in Alaska and Hawaii.

Back

Transportation Economics in the 21st Century - Commodity Transportation Survey Data

DATA LAST UPDATED February 2022

1. Commodity Transportation Survey, Dept. of Transportation

The CTS provides data on national freight flows in manufacturing. Data include quantities shipped and distances of shipments in tons between bilateral pairs of production areas in the United States. The definition of production areas varies across years. Data available for 1963, 1967, and 1972.

Provided by Stephen J. Redding.

- State-State Flows

2. Detailed 1963 Commodity Transportation Survey:

These files contain data on ton and ton-miles of shipments between census divisions and total shipments by mode and by distance. All the files have been digitized from tabulations published in the 1963 Commodity Transportation Survey (CTS).

Provided by Kerem Cosar.

- CTS1963

As used in: Cosar, Osotimehin and Popov, 2022, "Regional and Aggregate Implications of Transportation Costs", mimeo.

The Commodity Transportation Survey was superseded by the Commodity Flow Survey

Solution of the model's steady-state

- cost minimization + real wage equalization across regions $\rightarrow$ equilibrium prices
- Given prices $\rightarrow$ demand for intermediate goods, capital and labor
- Equilibrium allocation

$$Q = (I - F')^{-1} C^w$$

$$Q = (I - F')^{-1} \omega G' L$$

$$\underbrace{z(I - F')^{-1} \omega G' L}_{\text{labor demand}} = \underbrace{L}_{\text{labor supply}}$$

Solution is the eigenvector of $z(I - F')^{-1} \omega G$ associated with the unit eigenvalue.

Back

Iceberg results

Back

- Transportation margins no longer key determinant of gross output changes

	Baseline	Iceberg Transport Costs	
		same targets	same parameters
Transportation margin	1.596***	0.0463**	-0.0474*
	(0.000)	(0.032)	(0.053)
R^2	0.74	0.277	0.165

Notes: Sample includes only goods sectors (N = 72). p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- No change in the distance elasticity

$$\ln(X_{iodt}) = -1.454 \cdot \ln(dist) + 0.019 \cdot \ln(dist) \times \mathcal{I}(\text{t} = 2017) + \Gamma_{iodt} + \epsilon_{iodt}$$

$$p\text{-}value: \quad [0.000] \qquad [0.807]$$

Calibration: elasticities

Back

Substitution elasticities are $(x-1)/x$

Elasticity	**Parameter** (x)	elast.
btw intermediates	ρ	0.25
btw intermediate bundle and capital-labor	σ	0.7
btw final goods	χ	0.25
btw regional varieties	γ	8
Interest rate	r	0.04

Calibration: Targets (1/4)

	Value added		Dist. elasticity.		Out-of-region	
	data	model	data	model	data	model
Agriculture	0.014	0.029	2.0	2.0	1.8	1.7
Mining	0.024	0.030	3.1	2.9	0.6	0.6
Heavy manuf.	0.062	0.064	1.6	1.6	1.0	1.0
Light manuf.	0.070	0.055	1.2	1.2	1.6	1.6
Tradable services	0.346	0.359	-	-	-	-
Non-tradable services	0.454	0.433	-	-	-	-
Transp.	0.030	0.031	-	-	-	-
Transp. (va/gross output)	0.480	0.481	-	-	-	-

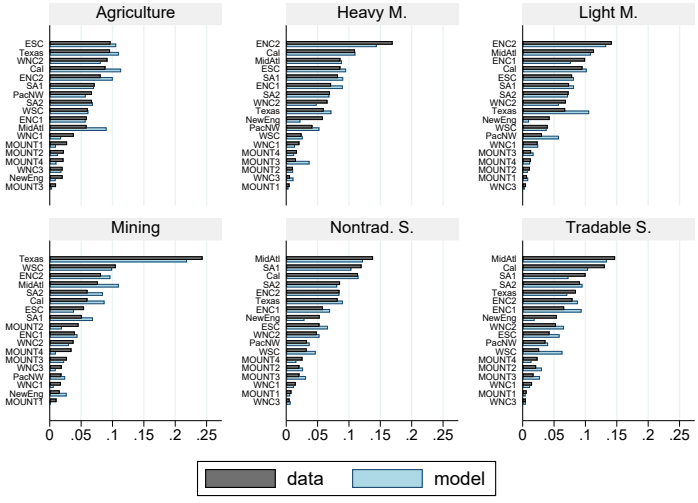
Calibration: Targets (2/4) Back

	Input-output shares	Emp. shares
Median abs. error	0.009	0.009

Calibration: Targets (3/4)

Back

Regional employment (as a share of sectoral employment)



Calibration: Targets (4/4)

Back

Input-output (as a share of the purchasing sector's gross output)

